

MAPLE: Macro–Micro Aligned Pigouvian Lane Control for Mixed-Autonomy Traffic Efficiency

Kexin Wang¹, Gavin Huang¹, Zehao Wang², Jiachen Li², and Ruolin Li^{1†}

Abstract—This paper proposes MAPLE, a closed-loop framework that leverages connected and automated vehicles (CAVs) for system-level congestion management at highway bottlenecks. By designing a lane-level Pigouvian price signal derived from real-time traffic states, MAPLE guides decentralized CAV lane-changing decisions without inter-vehicle communication or centralized coordination. We prove that this pricing mechanism induces an ε -approximate potential game, formally guaranteeing that individual CAV decisions align with the system-level congestion minimization objective by design, not by assumption. Simulation results across six mixed-autonomy scenarios demonstrate consistent and significant improvements over state-of-the-art baselines, with performance trends directly interpretable through the lens of the theoretical guarantee.

I. INTRODUCTION

Transportation systems worldwide are undergoing a profound transformation. As automated vehicle technologies mature and fleet-scale deployment accelerates [1], a new class of road user is emerging: connected and automated vehicles (CAVs) that are not merely passive participants in traffic, but controllable, communicable agents capable of coordination. Waymo’s commercial robotaxi service already delivers more than 400,000 weekly paid trips across multiple U.S. metropolitan cities and is expanding rapidly [2]. This trajectory marks the arrival of *mixed autonomy*: an era in which CAVs and human-driven vehicles (HDVs) share the same infrastructure, and in which the growing CAV population presents a unique opportunity. We can deliberately leverage these controllable agents as a distributed actuation layer for system-level traffic management, turning a technological milestone into a societal benefit.

Realizing this opportunity requires resolving a fundamental tension in traffic control. *Macroscopic* approaches, such as variable speed limits, ramp metering, and perimeter control, operate on aggregated traffic states and scale well to network-level analysis, issuing uniform control rules that guide population-level behavior [3]–[5]. However, because these rules are derived from aggregate conditions rather than individual vehicle states, they cannot adapt guidance to each vehicle’s specific situation. For instance, a CAV in a congested lane receives the same signal as one in a free-flowing lane, regardless of what action would be most beneficial for either. *Microscopic* approaches model or optimize individual vehicle decisions directly, enabling

fine-grained control of lane-changing and car-following behavior [6], [7], but without system-level awareness, locally optimal decisions can inadvertently worsen global traffic conditions. *Macro-micro* frameworks attempt to bridge these paradigms [8], [9], but introduce a deeper problem: *cross-layer incentive incompatibility*. System-level controllers prescribe lane allocations or guidance signals that individual vehicles have no incentive to follow [10], or that are feasible only under idealized assumptions of perfect information and full controllability [11]. More fundamentally, while the physical feedback between individual decisions and macroscopic traffic states is always present, existing frameworks do not formally characterize this feedback loop, leaving the alignment between individual incentives and system-level objectives unanalyzed and unguaranteed. What is missing is a framework that not only closes this loop architecturally, but formally proves the resulting individual decisions are aligned with the system objective by design.

In our prior work, we identified highway ramps as representative congestion bottlenecks [12], [13] and showed aggregate lane-choice behavior can be leveraged to improve system-level traffic performance [14]. Building on these findings, we propose **Macro-micro Aligned Pigouvian Lane control for mixed-autonomy traffic Efficiency (MAPLE)**, a closed-loop macro-micro framework instantiated on weaving ramps as a proof-of-concept bottleneck scenario.

MAPLE operates through a principled chain of mechanisms, illustrated in Fig. 1. The road geometry is divided into an upstream *control zone*, where CAVs execute lane-changing decisions, and a downstream *sensing zone*, where lane-level traffic states are collected by sensors. The system objective is to minimize total lane-level congestion cost, a cost whose lane-indexed structure captures inter-lane imbalance, making it naturally suited to guiding lane-changing decisions. Since this cost depends on the future macro state, which is unknown at decision time, we derive a closed-form physical perturbation model approximating how each CAV’s lane change alters lane density and speed. The gradient of this cost with respect to the lane-level macro state defines a *lane-based Pigouvian price signal*, broadcast by the macroscopic regulator to all upstream CAVs. Each CAV uses this signal to compute a local payoff measuring the congestion impact of its action, without requiring knowledge of other vehicles’ decisions. The resulting lane-changing actions collectively reshape the downstream macro state,

^{*1}University of Southern California, Los Angeles, CA, USA. Corresponding author email: kwang255@usc.edu

^{†2}University of California, Riverside, CA, USA.

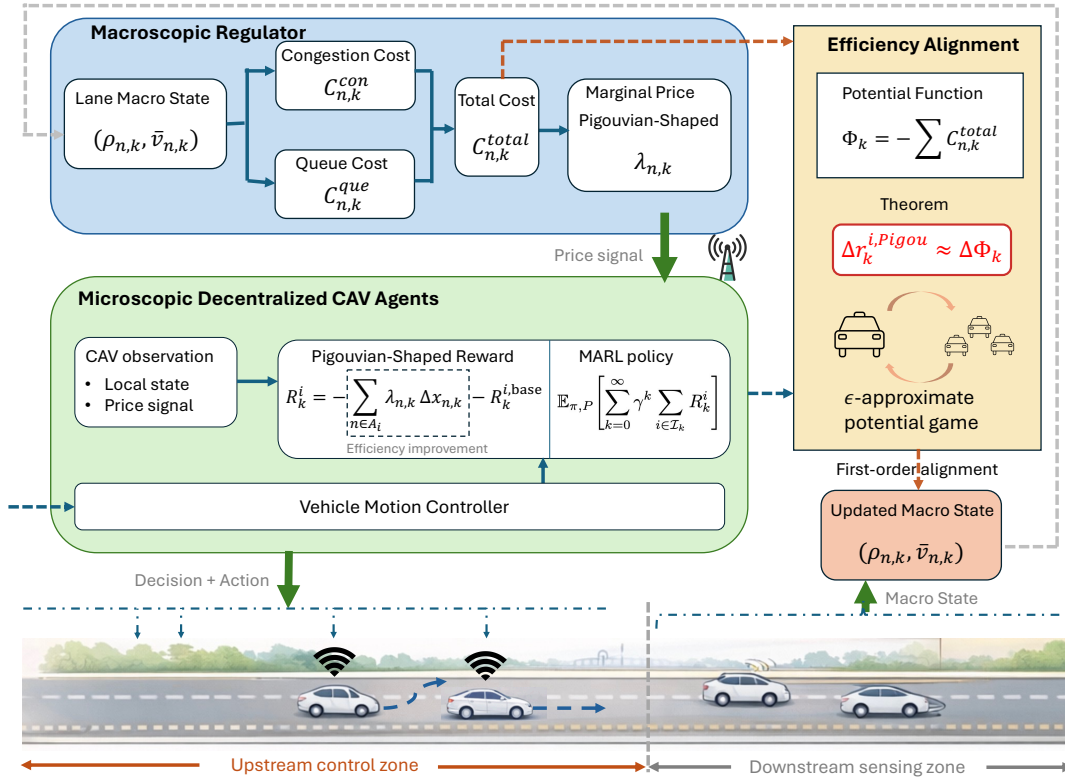


Fig. 1: This figure illustrates MAPLE framework. The macroscopic traffic state is sensed and collected from the downstream evolution region, and the macroscopic regulator uses the information to generate price signals. Microscopic decentralized CAV agents combine these signals with a local control policy (e.g. MARL) to make and execute lane-change decisions in the upstream decision region. The resulting actions propagate downstream and update the measured macro state, forming a micro $\leftrightarrow$ macro feedback loop.

which in turn updates future price signals, forming a closed loop between macroscopic sensing and microscopic control.

The central theoretical contribution is to show this pricing mechanism induces the CAV interaction to form an ϵ -approximate potential game at each time step: any unilateral improvement in an individual CAV's payoff corresponds to a reduction in total congestion cost, up to a bounded and characterized error. Individual rationality and collective efficiency are therefore aligned by design, not by assumption. This provides the principled foundation for embedding the Pigouvian payoff into a multi-agent reinforcement learning (MARL) reward design, enabling CAVs to learn effective lane-changing policies through decentralized execution.

The main contributions of this paper are threefold. First, we propose a closed-loop macro-micro control framework in which a macroscopic regulator broadcasts lane-level Pigouvian price signals derived from real-time sensor measurements, while decentralized CAVs respond via lane-changing decisions that propagate downstream and update the macro state, formally closing the feedback loop between macroscopic sensing and microscopic control. Second, we prove that the Pigouvian payoff induces an ϵ -approximate potential game at each time step, with a time-uniform error bound under moderate-density conditions, establishing that decentralized CAV decisions are aligned with the system-level

congestion minimization objective by design, not by assumption. Third, simulation results across six traffic scenarios demonstrate consistent improvements over rule-based and learning-based baselines, with performance trends directly interpretable through the lens of the theoretical guarantee.

II. PROBLEM FORMULATION

A. Zone Division

We consider a highway weaving ramp as the pilot scenario for the proposed framework. As illustrated in Fig. 2, the road geometry is divided into two functionally distinct zones that define the macro-micro architecture of MAPLE.

The *upstream control zone* is the region preceding the weaving ramp, where CAVs observe their local surroundings and execute lane-changing decisions. This zone is the locus of microscopic control: individual vehicle actions are taken here, and it is where the Pigouvian price signals broadcast by the macroscopic regulator are received and acted upon.

The *downstream sensing zone* corresponds to the weaving ramp itself, where macroscopic traffic states are continuously collected via sensors. Lane-level density and average speed are measured at discrete time steps. This zone is the locus of macroscopic sensing: it captures the downstream consequences of upstream lane-changing decisions and feeds them back to the regulator.

These two zones are spatially separated but dynamically coupled: decisions made in the control zone propagate downstream and alter the macro state in the sensing zone, while measurements from the sensing zone are used to compute price signals that guide future decisions in the control zone. This bidirectional coupling is the structural basis of the closed-loop feedback mechanism at the heart of MAPLE, and it distinguishes the framework from prior hierarchical designs in which information flows only top-down.

B. Game Setting

In a mixed-autonomy traffic system, both CAVs and HDVs pursue their own travel objectives, and their collective lane-changing behavior determines the evolution of downstream congestion. The fundamental challenge is that congestion is a *shared externality*: each vehicle's lane choice imposes costs on all other road users, yet no individual vehicle has an incentive to internalize these costs without deliberate mechanism design. For CAVs, which are controllable and communicable, this creates a tractable intervention point: if their lane-changing decisions can be guided by a system-designed payoff signal, it becomes possible to steer collective behavior toward socially efficient outcomes.

At time step k , the macroscopic traffic state of the downstream sensing zone is $x_k = (x_{1,k}, \dots, x_{N,k})$, where $x_{n,k} = (\rho_{n,k}, \bar{v}_{n,k})$ denotes the density and average speed of lane $n \in \mathcal{N} = \{1, \dots, N\}$. The system objective is to find a joint lane-changing action $a_k = (a_k^i)_{i \in \mathcal{I}_k}$ over all active CAVs $\mathcal{I}_k$ that minimizes total downstream congestion cost at the resulting future macro state $\tilde{x}_k$:

$$\min_{a_k \in \mathcal{A}_k} \sum_{n=1}^N C_{n,k}^{\text{total}}(\tilde{x}_{n,k}(a_k)), \quad (1)$$

where $C_{n,k}^{\text{total}}$ is lane-level congestion cost defined in (5), $\mathcal{A}_k^i \subseteq \{1, \dots, N\}$ is the set of feasible target lanes for CAV i , and $\mathcal{A}_k := \prod_{i \in \mathcal{I}_k} \mathcal{A}_k^i$ is the joint action space.

Directly solving (1) requires centralized coordination over a large distributed population and accurate knowledge of the future state $\tilde{x}_k$, neither of which can be easily attained in real time. A game-theoretic formulation is therefore the natural framework: rather than solving for a centralized optimal action, we design a local payoff r_k^i for each CAV such that decentralized self-interested optimization collectively approximates (1). To formalize this interaction, we model the joint decision-making of CAVs at each time step as a *stage game*. Intuitively, a stage game captures a one-shot strategic interaction: each participant chooses an action, and each receives an evaluation score that depends on what everyone chose. Here, participants are the active CAVs $\mathcal{I}_k$, their actions are lane-changing decisions $\mathcal{A}_k^i$, and their evaluation scores are payoffs r_k^i that reflect how much their action helps or hurts downstream congestion. The game is parameterized by the current macro state x_k , meaning the set of players, their feasible actions, and their payoff evaluations are all defined relative to the traffic condition at time step k . This

is compactly written as:

$$\mathcal{G}_k(x_k) = (\mathcal{I}_k, \{\mathcal{A}_k^i\}_{i \in \mathcal{I}_k}, \{r_k^i\}_{i \in \mathcal{I}_k}), \quad (2)$$

where r_k^i 's construction is deferred to Section IV-A, and reflects the system impact: how much does CAV i 's lane change reduce total downstream congestion relative to staying in current lane? A positive r_k^i means the action helped the system; a negative means it made congestion worse.

Remark 1 (Only CAVs' actions subject to control). HDVs are also players in the traffic system, pursuing their own travel objectives through lane-changing and car-following behavior. However, unlike CAVs, HDVs are not subject to external payoff design and cannot be directly actuated. We therefore treat HDV behavior as part of the stochastic environment dynamics, a standard modeling convention in mixed-autonomy traffic control [11], [15]. The stage game $\mathcal{G}_k$ thus governs the controllable CAV population only, while HDV interactions are captured implicitly through the macro state x_k and the environment transition kernel in the MARL experiments (Section II-C).

Therefore, the central research question this paper addresses is: *can r_k^i be designed such that decentralized CAV decisions are formally aligned with the system objective (1)?* We answer this affirmatively by showing that a Pigouvian pricing mechanism induces $\mathcal{G}_k$ to form an ε -approximate potential game, guaranteeing that individual reward improvements correspond to reductions in total congestion cost up to a bounded error.

C. MARL Instantiation

The stage game $\mathcal{G}_k$ characterizes the per-step decision structure but does not address how CAVs learn policies over time, or how individual safety is enforced during execution. We therefore embed $\mathcal{G}_k$ into a Markov game $\mathcal{M} = (\mathcal{I}, \mathcal{S}, \{\mathcal{O}^i\}_{i \in \mathcal{I}}, \{\mathcal{A}^i\}_{i \in \mathcal{I}}, P, \{R^i\}_{i \in \mathcal{I}}, \gamma)$, where $\mathcal{A}_k^i \subseteq \mathcal{A}^i$ is the time-varying feasible action subset at step k , and only CAVs currently in the control zone, $\mathcal{I}_k \subseteq \mathcal{I}$, participate at each step. Each active CAV $i \in \mathcal{I}_k$ receives an augmented observation $z_k^i = (o_k^i, x_k)$, combining local state information o_k^i , indicating own speed, current lane, and nearby vehicle positions, with the broadcast macro state x_k , and selects a target-lane action $a_k^i \in \mathcal{A}_k^i$ via policy $\pi^i(a_k^i | z_k^i)$. The reward R_k^i consists of a base term enforcing local safety and comfort constraints, and a congestion signal derived from downstream traffic states that guides CAVs toward system-level efficiency, formally constructed in Section IV-A. The joint policy $\pi = (\pi^i)_{i \in \mathcal{I}}$ is learned under centralized training and decentralized execution by maximizing $\max_{\pi} \mathbb{E}_{\pi, P} [\sum_{k=0}^{\infty} \gamma^k \sum_{i \in \mathcal{I}_k} R_k^i]$.

III. TRACTABLE CONGESTION COST FORMULATION

The system objective (1) minimizes the sum of lane-level congestion costs $\sum_n C_{n,k}^{\text{total}}$ across all lanes. This lane-indexed structure is not incidental, and it is precisely what makes the cost function appropriate for a lane-changing control problem. Unlike aggregate traffic metrics that collapse

all lanes into a single flow or density value, the lane-level decomposition captures *imbalance across lanes*: a scenario where one lane is severely overloaded while an adjacent lane is underutilized incurs a higher cost than one where the same total flow is evenly distributed. This sensitivity to inter-lane imbalance is what gives the Pigouvian price signal its directional power, i.e., it can tell a CAV not just that the system is congested, but *which lane* it should move to in order to reduce that imbalance.

However, evaluating $C_{n,k}^{\text{total}}$ requires knowing the future macro state $\tilde{x}_k$, which is unknown at decision time k . This section resolves that challenge in two steps. First, we derive a closed-form perturbation model that approximates $\tilde{x}_{n,k}$ from current observables and the planned action a_k . Second, we construct $C_{n,k}^{\text{total}}$ as a closed-form, differentiable function of the current macro state x_k .

A. Action-induced State Evolution

Since each CAV's lane-changing action a_k^i induces a perturbation in the macro state, which can be represented by the changes in $\Delta\rho_{n,k}^i$ and $\Delta\bar{v}_{n,k}^i$. Therefore, we employ a simplified physical model to characterize these perturbations. This abstraction is introduced to maintain analytical tractability. In practical, the proposed formulation can be extended to more detailed traffic models with the same physical interpretation.

The density perturbation $\Delta\rho_{n,k}^i$ follows directly from traffic density definition. Specifically, when a vehicle enters or leaves a downstream lane of length L_n , the resulting change in density at time step k is given by:

$$\Delta\rho_{n,k}^i = \frac{\epsilon_i \sigma_{n,k}^i}{L_n}, \quad (3)$$

where L_n denotes the effective length of downstream region in lane n , and ϵ_i represents the effective length of agent i . For homogeneous traffic, we normalize $\epsilon_i = 1$ for all CAVs. The indicator $\sigma_{n,k}^i \in \{-1, 0, 1\}$ captures the directional effect of CAV's action: $\sigma_{n,k}^i = 1$ if it enters lane n , $\sigma_{n,k}^i = -1$ if it leaves lane n , and $\sigma_{n,k}^i = 0$ for all other lanes.

The speed perturbation $\Delta\bar{v}_{n,k}^i$ is derived from the update of the downstream average speed at lane n with $M_{n,k} = \rho_{n,k} L_n$ vehicles. When entering a lane, the speed perturbation is exactly $\frac{v_k^i - \bar{v}_{n,k}}{M_{n,k} + 1}$; when leaving, it is $\frac{v_k^i - \bar{v}_{n,k}}{M_{n,k} - 1}$. Under the assumption that $M_{n,k} \gg 1$, which is typical in congested downstream regions, $\Delta\bar{v}_{n,k}^i$ can be approximated by:

$$\Delta\bar{v}_{n,k}^i = \frac{\sigma_{n,k}^i (v_k^i - \bar{v}_{n,k})}{\rho_{n,k} L_n}. \quad (4)$$

Hence joint actions of all CAVs determine the evolution of the future macroscopic state $\tilde{x}_{n,k}$. Moreover, since each CAV contributes an additive state variation $\Delta x_{n,k}^i$, the overall state transition can be written as $\tilde{x}_{n,k} := x_{n,k} + \sum_i \Delta x_{n,k}^i$. With $\tilde{x}_k$ expressed as a closed-form function of the current state and the joint action, the congestion cost $C_{n,k}^{\text{total}}(\tilde{x}_{n,k})$ can now be written explicitly in terms of current observables. We construct this cost from two complementary components,

each capturing a distinct dimension of congestion at the weaving ramp.

B. Congestion cost formulation

To operationalize the directional guidance motivated above, we construct a lane-indexed cost $C_{n,k}^{\text{total}}$ that penalizes both current downstream congestion and predicted upstream queue buildup independently per lane. This lane-differentiated structure is precisely what gives the resulting price signal its directional power: it tells a CAV not just that the system is congested, but which lane to move to in order to reduce inter-lane imbalance. Formally, we define

$$C_{n,k}^{\text{total}} := \zeta_1 C_{n,k}^{\text{down}} + \zeta_2 C_{n,k}^{\text{up}}, \quad (5)$$

where $\zeta_1, \zeta_2 \geq 0$ with $\zeta_1 + \zeta_2 = 1$ balance the contribution of current and predicted congestion.

Downstream congestion cost. Since the BPR function captures congestion-induced travel delay, we use it to define the downstream congestion cost following classical traffic flow theory [16]:

$$C_{n,k}^{\text{down}} = t_{n,k}^0 \left[\rho_{n,k} \bar{v}_{n,k} + \alpha \frac{(\rho_{n,k} \bar{v}_{n,k})^{\beta+1}}{C_n^\beta} \right], \quad (6)$$

where $t_n^0 = \frac{L_n}{v_n^{\text{free}}}$ is the free-flow travel time, C_n is the lane capacity, and α, β are standard BPR parameters.

Upstream congestion cost. Relying solely on $C_{n,k}^{\text{down}}$ overlooks how upstream lane-changing decisions propagate into future congestion. To capture this effect, we introduce $C_{n,k}^{\text{up}}$ based on the predicted net vehicle accumulation over one control interval. Under the assumption that traffic flow is relatively stable within each stage, the future congestion trend can be inferred from the imbalance between the upstream inflow $Q_{n,k}^{\text{in}}$, measured by CAVs, and the downstream inflow $q_{n,k}^{\text{in}}$, estimated via the cell transmission model (CTM) [17].

As illustrated in Figure 2, $q_{n,k}^{\text{in}}$ is inferred from flow evolution within the downstream region. Since CTM evaluates flow and density at a finite number of intermediate points [18], we divide the weaving ramp into four segments $s \in \{1, 2, 3, 4\}$, yielding:

$$q_{n,k}^{\text{in}} = q_{n,k}^{\text{out}} - f_{n,k}^{\text{on}} + f_{n,k}^{\text{off}} + \frac{1}{\Delta t_c} \sum_{s=1}^4 L_n^s (\rho_{n,k+1}^s - \rho_{n,k}^s), \quad (7)$$

where $f_{n,k}^{\text{on}}$ and $f_{n,k}^{\text{off}}$ denote on-ramp and off-ramp flows, and L_n^s is the effective length of segment s . All terms on the right-hand side are obtainable from downstream sensors, where $\rho_{n,k+1}^s$ is approximated by a one-step delayed estimate as detailed in Remark 2. A sketch proof transferring this from the standard CTM is provided in Appendix VII-A.

Remark 2. Since $\rho_{n,k+1}^s$ depends on a future time step, prices are computed using a one-step delayed estimate, i.e., replacing $\rho_{n,k+1}^s - \rho_{n,k}^s$ with $\rho_{n,k}^s - \rho_{n,k-1}^s$.

The upstream congestion cost is then defined as:

$$C_{n,k}^{\text{up}} := (Q_{n,k}^{\text{in}} - q_{n,k}^{\text{in}}) \Delta t_c. \quad (8)$$

A positive $C_{n,k}^{\text{up}}$ indicates net vehicle accumulation and rising congestion; a negative value indicates queue dissipation.

Manuscript 1901 submitted to 2026 65th IEEE Conference on
Decision and Control (CDC). Received March 31, 2026.

- 1) **Bounded regime:** $\rho_{n,k} \geq \rho_{\min} > 0$ and $L_n \geq L_{\min} > 0$, where $\rho_{\min}$ and $L_{\min}$ are respective lower bounds;
- 2) **Bounded velocity deviation:** $|v_k^i - \bar{v}_{n,k}| \leq V$ for all i ;
- 3) **Smooth potential:** Φ_k is continuously differentiable with Lipschitz gradient constant Lip_k .

Then the deviation between the Pigou-based payoff difference and the potential function difference is bounded:

$$\left| (r_k^{i,\text{Pigou}}(a_k^{i'}) - r_k^{i,\text{Pigou}}(a_k^{i''})) - (\Phi_k(x_k^{i'}) - \Phi_k(x_k^{i''})) \right| \leq \epsilon_k^i, \quad (17)$$

where $\epsilon_k^i \leq \frac{2 \text{Lip}_k}{L_{\min}^2} \left(1 + \frac{V^2}{\rho_{\min}^2} \right)$.

Consequently, setting $\epsilon_k = \max_i \epsilon_k^i$, the stage game $\mathcal{G}_k$ is an ϵ_k -approximate potential game.

The sketch of the proof is provided in Appendix VII-C. From Equation (17), it follows that ϵ_k decreases as the segment length L_n increases and the minimum density $\rho_{\min}$ becomes larger. This result indicates that the proposed Pigou-based framework is more effective in longer road segments and under moderate traffic density conditions.

Remark 3 (Moderate-density operating regime). The proposed perturbation model is intended for traffic regimes with moderate density, where lane-changing interactions are active and have a non-negligible impact on the macroscopic state. In extremely low-density conditions, vehicle interactions are sparse and lane-changing behavior becomes negligible.

Proposition 1 (Existence of pure ϵ_k -Nash equilibrium). *At time step k , each action set $\mathcal{A}_k^i$ is finite. Let joint action $a_k^* = (a_k^{i,*})_{i \in \mathcal{I}_k}$ and*

$$a_k^* \in \arg \max_{a_k \in \mathcal{A}_k} \Phi_k(\tilde{x}_k), \quad (18)$$

where $\mathcal{A}_k := \prod_{i \in \mathcal{I}_k} \mathcal{A}_k^i$. Then a_k^* is a pure ϵ_k -Nash equilibrium of the stage game $\mathcal{G}_k$. Moreover, any maximizer of Φ_k equivalently minimizes the total congestion cost:

$$C_k^{\text{total}}(\tilde{x}_k(a_k^*)) = \min_{a_k \in \mathcal{A}_k} C_k^{\text{total}}(\tilde{x}_k). \quad (19)$$

The sketch proof is shown in Appendix VII-D. By definition of the potential function, maximizing Φ_k is equivalent to minimizing the total congestion cost. Combined with Theorem VII-C, this implies that individual best responses are aligned up to bounded error, with the system-level objective.

This ϵ -approximate potential game provides a principled foundation for learning-based implementations. Therefore, we further demonstrate how MAPLE is incorporated as a modular into dynamic learning schemes via reward design to improve system efficiency and mitigate bottleneck congestion in next section.

V. NUMERICAL EXPERIMENTS

A. Scenario Construction and Training Setting

The experiments are implemented in Python with SUMO as the traffic simulator. Each training episode lasts 80 seconds, and the MARL policy is trained for 1.5 million steps using Multi-Agent Proximal Policy Optimization

TABLE I: Scenario settings with different demand levels and CAV penetration rates.

Scenario	Demand (veh/hr)	CAV Penetration (%)
Scenario I	7,200	50
Scenario II	12,000	30
Scenario III	12,000	50
Scenario IV	12,000	80
Scenario V	15,600	50
Scenario VI	18,000	50

(MAPPO) [20] under a centralized training and decentralized execution paradigm. We evaluate the proposed method under six scenarios with varying demand levels and CAV penetration rates, as summarized in Table I.

B. Performance Evaluation

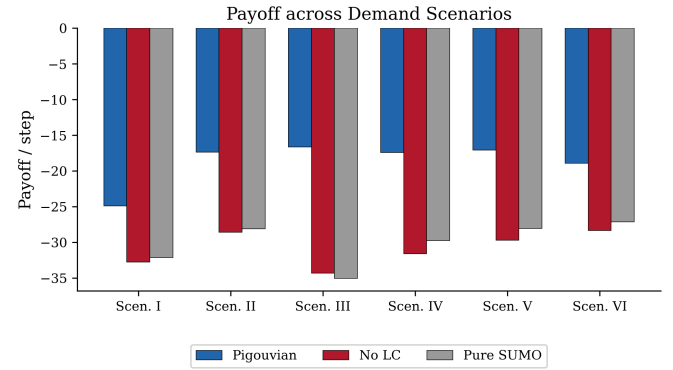


Fig. 3: Average payoff per step across six scenarios, comparing the Pigou-shaped framework against other baselines.

In Table II, we compare the proposed Pigou-shaped framework against six baselines: No Lane Change (No LC), Pure SUMO cooperative control, Downstream-Congestion-only reward, Upstream-Congestion-only reward, and two rule-based strategies (periodic and gap-based). The system performance is evaluated across five metrics: individual payoff per step, total congestion cost, throughput (veh/s), average speed (km/h), and average travel time (s).

The proposed framework achieves the highest payoff per step (-14.71) and the lowest congestion cost (13.21) under Scenario III, corresponding to a 50% improvement in payoff and a 52.5% reduction in congestion cost relative to the No LC baseline. Travel time is also reduced by 6.5%. Compared with the congestion-only and queue-only ablations, the proposed framework better balances downstream congestion and upstream spillback, confirming that jointly optimizing $C_{n,k}^{\text{total}}$ outperforms single-objective designs. While the Pigouvian strategy yields lower average speeds than some baselines, this reflects an expected speed-throughput tradeoff: targeted lane changes redistribute traffic density and create gaps in the flow, reducing congestion even as they temporarily slow individual vehicles.

TABLE II: Baseline comparison on Scenario III (100% demand, 9 600 veh/h, 50% CAV penetration). Metrics averaged over 5 episodes of 80 s each. Higher payoff and lower congestion cost indicate better performance.

Strategy	Payoff/step	Cong. Cost	Throughput (veh/s)	Avg Speed (km/h)	Avg Travel Time (s)
MAPLE (ours)	-14.71	13.21	0.065	35.4	29.2
Downstream-only (ablation)	-18.68	17.15	0.076	38.3	29.6
Upstream-only (ablation)	-17.96	16.43	0.078	37.8	29.9
Rule-based (periodic)	-23.80	22.25	0.082	35.8	31.1
Rule-based (gap)	-24.66	23.12	0.080	33.9	30.5
No Lane Change (LC)	-29.40	27.83	0.114	40.6	31.3
Pure SUMO	-28.55	26.99	0.108	41.0	30.6

Pigouvian vs. Baseline: payoff **+50.0%**, congestion **+52.5%**, travel time +6.5%. Ablation vs. Baseline: downstream-only +36.5%, upstream-only +38.9%.

C. Robustness to Demand Variations

A key question is whether the proposed framework maintains consistent improvement across varying traffic conditions. We evaluate performance across six scenarios spanning different demand levels and CAV penetration rates in Figure 3. Since the primary purpose of this figure is to highlight how performance varies with traffic demand and CAV penetration, we select No LC and Pure SUMO as reference points for their visual contrast.

The Pigouvian strategy achieves the strongest gains under Scenario III (medium demand, 50% CAV penetration), which corresponds to the moderate-density regime assumed in Theorem 1. Under low-demand conditions (Scenario I), the benefit of lane changes diminishes, as vehicles are sufficiently sparse that density redistribution has limited effect on congestion. Under peak-demand conditions (Scenario VI), lane-change opportunities are restricted by high vehicle density, reducing the scope for active control. Both observations are consistent with the theoretical error bound in Equation 17, which predicts larger approximation error outside the moderate density regime.

VI. CONCLUSION

This paper proposed MAPLE, a closed-loop macro-micro framework for congestion mitigation at highway bottlenecks in mixed-autonomy traffic. By designing a lane-level Pigouvian pricing mechanism derived from real-time traffic states, MAPLE guides decentralized CAV lane-changing decisions and formally guarantees via an ε -approximate potential game, that individual decisions align with system-level congestion minimization by design. Simulation results across six scenarios confirm consistent improvements over baselines, with the largest gains under medium-to-high CAV penetration rates, directly interpretable through the theoretical guarantee. Future extensions include modeling HDV strategic behavior within the incentive mechanism and generalizing the stage game formulation to a dynamic game setting to capture temporal coupling across decision stages.

VII. APPENDIX

A. CTM proof

For the highway segment s without weaving ramps, vehicle conservation in each lane n at time step k can be written

$$\text{as [17]: } \rho_{n,k+1}^s = \rho_{n,k}^s + \frac{\Delta t}{L_n^s} (q_{n,k}^s - q_{n,k}^{s+1}).$$

For the 4-cell segment with one on-ramp at cell 2 and one off-ramp at cell 3 in Figure 2, the per-lane conservation equations are: cell 2: $\rho_{n,k+1}^2 = \rho_{n,k}^2 + \frac{\Delta t}{L_n^2} (q_{n,k}^2 + r_{n,k}^{\text{on}} - q_{n,k}^3)$, and cell 3: $\rho_{n,k+1}^3 = \rho_{n,k}^3 + \frac{\Delta t}{L_n^3} (q_{n,k}^3 + f_{n,k}^{\text{off}} - r_{n,k}^{\text{off}})$, where $r_{n,k}^2$ and $f_{n,k}^3$ are traffic flow that enter and leave the weaving ramp. Then we solve for $q_{n,k}^{\text{in}}$: $q_{n,k}^{\text{in}} = q_{n,k}^{\text{out}} - r_{n,k}^{\text{on}} + f_{n,k}^{\text{off}} + \frac{1}{\Delta t} \sum_{i=1}^4 L_n^i (\rho_{n,k+1}^i - \rho_{n,k}^i)$.

B. Convexity, Monotonicity, and Smoothness

Downstream congestion cost $C_{n,k}^{\text{down}}$:

$$\frac{dC_{n,k}^{\text{down}}}{d\rho_{n,k}} = \bar{v}_{n,k} t_n^0 [1 + \alpha(1 + \beta) \frac{(\rho_{n,k} \bar{v}_{n,k})^\beta}{C_n}] \geq 0, \quad (20)$$

$$\frac{d^2 C_{n,k}^{\text{down}}}{d\rho_{n,k}^2} = t_n^0 [\alpha\beta(1 + \beta) \frac{\bar{v}_{n,k}^2 (\rho_{n,k} \bar{v}_{n,k})^{\beta-1}}{C_n^\beta}] \geq 0. \quad (21)$$

Since $\alpha, \beta, \rho_{n,k}, \bar{v}_{n,k} \geq 0$, this shows $C_{n,k}^{\text{down}}$ is convex and nondecreasing as a function of $\rho_{n,k}$.

$$\frac{dC_{n,k}^{\text{down}}}{d\bar{v}_{n,k}} = \rho_{n,k} t_n^0 [1 + \alpha(1 + \beta) \frac{(\rho_{n,k} \bar{v}_{n,k})^\beta}{C_n}] \geq 0, \quad (22)$$

$$\frac{d^2 C_{n,k}^{\text{down}}}{d\bar{v}_{n,k}^2} = t_n^0 [\alpha\beta(1 + \beta) \frac{\rho_{n,k}^2 (\rho_{n,k} \bar{v}_{n,k})^{\beta-1}}{C_n^\beta}] \geq 0. \quad (23)$$

Since $\alpha, \beta, \rho_{n,k}, \bar{v}_{n,k} \geq 0$, this shows $C_{n,k}^{\text{down}}$ is convex and nondecreasing as a function of $\bar{v}_{n,k}$.

Assume $\rho_{n,k}$ and $\bar{v}_{n,k}$ are bounded by $\rho_{n,k}^{\max}$ and $\bar{v}_{n,k}$, then Equation (21) and (23) are bounded, which shows $C_{n,k}^{\text{down}}$ is smooth for both $\rho_{n,k}$ and $\bar{v}_{n,k}$.

Upstream congestion cost $C_{n,k}^{\text{up}}$

From Equation (8) and Remark 2, since in $\tilde{G}_k$ $\rho_{n,k-1}$ happened at the last time step $k-1$, which does not change based on current time step k , thus we treat it as a constant. Also, $q_{n,k}^{\text{out}} := \rho_{n,k} \bar{v}_{n,k}$ due to $q_{n,k}^{\text{out}}$ is the downstream outflow.

$$\frac{dC_{n,k}^{\text{up}}}{d\rho_{n,k}} = -\Delta t_c (\bar{v}_{n,k} + \frac{L_n}{\Delta t_c}) \leq 0, \quad \frac{d^2 C_{n,k}^{\text{up}}}{d\rho_{n,k}^2} = 0. \quad (24)$$

which shows $C_{n,k}^{\text{up}}$ is convex and nonincreasing for $\rho_{n,k}$.

$$\frac{dC_{n,k}^{\text{up}}}{d\bar{v}_{n,k}} = -\Delta t_c \rho_{n,k} \leq 0, \quad \frac{d^2 C_{n,k}^{\text{up}}}{d\bar{v}_{n,k}^2} = 0. \quad (25)$$

Hence $C_{n,k}^{\text{up}}$ is convex and nonincreasing for $\bar{v}_{n,k}$. Since

Equation (24) and (25) are bounded, it shows $C_{n,k}^{\text{up}}$ is smooth for both $\rho_{n,k}$ and $\bar{v}_{n,k}$.

Total congestion cost $C_{n,k}^{\text{total}}$: From Equation (5), $C_{n,k}^{\text{total}}$ equals sum of weighted $C_{n,k}^{\text{down}}$ and $C_{n,k}^{\text{up}}$. Since $C_{n,k}^{\text{down}}$ and $C_{n,k}^{\text{up}}$ are continuously differentiable on any bounded feasible domain. $C_{n,k}^{\text{total}}$ is smooth with respect to $\rho_{n,k}$ and $\bar{v}_{n,k}$.

C. Theorem 1 proof

Proof. Fix a time step k and an agent $i \in \mathcal{I}_k$. Consider two unilateral actions $a_k^{i'}, a_k^{i''} \in A_k^i$, while the actions of all other agents are fixed. The corresponding perturbed macro states are $x_k' = x_k + \Delta x_k^i(a_k^{i'})$ and $x_k'' = x_k + \Delta x_k^i(a_k^{i''})$.

Since $C_{n,k}^{\text{total}}$ is continuously differentiable with bounded second-order partial derivatives on the relevant bounded domain, Φ_k has a Lipschitz continuous gradient on that domain. Therefore, there exists a constant $Lip_k > 0$ such that for any perturbation Δx_k^i in the feasible domain,

$$|\Phi_k(x_k + \Delta x_k^i) - \Phi_k(x_k) - \nabla_x \Phi_k(x_k)^\top \Delta x_k^i| \leq \frac{Lip_k}{2} \|\Delta x_k^i\|^2. \quad (26)$$

Applying (26) to $\Delta x_k^i(a_k^{i'})$ and $\Delta x_k^i(a_k^{i''})$, and the sign convention of the payoff is chosen to align individual improvement with a decrease in total congestion cost, we obtain

$$\begin{aligned} & |(r_k^{i, \text{Pigou}}(a_k^{i'}) - r_k^{i, \text{Pigou}}(a_k^{i''})) - (\Phi_k(x_k') - \Phi_k(x_k''))| \\ & \leq \frac{Lip_k}{2} (\|\Delta x_k^i(a_k^{i'})\|^2 + \|\Delta x_k^i(a_k^{i''})\|^2) = \epsilon_k^i. \end{aligned} \quad (27)$$

From Equation (3) and (4), and the boundary $\rho_{\min}, L_{\min}, V$, we have: $|\Delta \rho_{n,k}^i| \leq \frac{1}{L_{\min}}$, $|\Delta \bar{v}_{n,k}^i| \leq \frac{V}{\rho_{\min} L_{\min}}$. Hence: $\|\Delta x_k^i(a_k^i)\|^2 = \sum_{n \in \mathcal{A}_i} (|\Delta \rho_{n,k}^i|^2 + |\Delta \bar{v}_{n,k}^i|^2) \leq \frac{2}{L_{\min}^2} (1 + \frac{V^2}{\rho_{\min}^2})$. Then we have the boundary for every $\epsilon_k^i \leq \frac{2Lip_k}{L_{\min}^2} (1 + \frac{V^2}{\rho_{\min}^2})$. Thus $\epsilon_k = \max_i \epsilon_k^i \leq \frac{2Lip_k}{L_{\min}^2} (1 + \frac{V^2}{\rho_{\min}^2})$. $\mathcal{G}_k$ is an ϵ_k -approximate potential game with bounded error. $\square$

D. Proposition 1 Proof

Proof. Since each action set $\mathcal{A}_k^i$ is finite, the joint action space $\mathcal{A}_k = \prod_{i \in \mathcal{I}_k} \mathcal{A}_k^i$ is finite. Hence there exists at least one joint action profile satisfy (18).

Suppose a_k^* is not a pure ϵ -Nash equilibrium. Then there exists at least one CAV i such that

$$r_k^{i, \text{Pigou}}(a_k^i, a_k^{-i,*}) > r_k^{i, \text{Pigou}}(a_k^{i,*}, a_k^{-i,*}) + \epsilon_k.$$

By Theorem 1, $\mathcal{G}_k$ is an ϵ -approximate potential game, so for the same unilateral deviation,

$$\Phi_k(x_k(a_k^i, a_k^{-i,*})) > \Phi_k(x_k(a_k^{i,*}, a_k^{-i,*})).$$

This contradicts the definition of a_k^* as a maximizer of Φ_k over the joint action space. Therefore, no player can improve its payoff by more than ϵ through unilateral deviation, and a_k^* is a pure ϵ -Nash equilibrium.

Finally, since $\Phi_k(x_k) = -C_k^{\text{total}}(x_k)$, maximizing Φ_k is equivalent to minimizing the total congestion cost. Hence

$$C_k^{\text{total}}(x_k(a_k^*)) = \min_{a_k \in \mathcal{A}_k} C_k^{\text{total}}(x_k(a_k)).$$

$\square$

REFERENCES

- [1] U.S. Department of Transportation, "Automated vehicles comprehensive plan," 2021.
- [2] Associated Press, "Waymo's robotaxis expand to 10 major u.s. markets with continued growth," 2026.
- [3] W.-L. Jin, "A kinematic wave theory of lane-changing traffic flow," *Transportation research part B: methodological*, vol. 44, no. 8-9, pp. 1001-1021, 2010.
- [4] X.-Y. Lu, P. Varaiya, R. Horowitz, D. Su, and S. E. Shladover, "A new approach for combined freeway variable speed limits and coordinated ramp metering," in *13th International IEEE Conference on Intelligent Transportation Systems*. IEEE, 2010, pp. 491-498.
- [5] Q. Guo and X. J. Ban, "Macroscopic fundamental diagram based perimeter control considering dynamic user equilibrium," *Transportation Research Part B: Methodological*, vol. 136, pp. 87-109, 2020.
- [6] Z. Wang, H. Huang, J. Tang, X. Meng, and L. Hu, "Velocity control in car-following behavior with autonomous vehicles using reinforcement learning," *Accident Analysis & Prevention*, vol. 174, p. 106729, 2022.
- [7] E. Vinitsky, A. Kreidieh, L. Le Flem, N. Kheterpal, K. Jang, C. Wu, F. Wu, R. Liaw, E. Liang, and A. M. Bayen, "Benchmarks for reinforcement learning in mixed-autonomy traffic," in *Conference on robot learning*. PMLR, 2018, pp. 399-409.
- [8] W. Lu, Z. Yi, B. Ran, and H. Yang, "Refined synergistic optimization control method for mitigating congestion near bottleneck via deep reinforcement learning in a connected environment," *Transportation Research Part C: Emerging Technologies*, vol. 185, p. 105582, 2026.
- [9] I. Papamichail, A. Kotsialos, I. Margonis, and M. Papageorgiou, "Coordinated ramp metering for freeway networks—a model-predictive hierarchical control approach," *Transportation Research Part C: Emerging Technologies*, vol. 18, no. 3, pp. 311-331, 2010.
- [10] R. W. Hall and D. Lottspeich, "Optimized lane assignment on an automated highway," *Transportation Research Part C: Emerging Technologies*, vol. 4, no. 4, pp. 211-229, 1996.
- [11] N. Mehr and R. Horowitz, "Pricing traffic networks with mixed vehicle autonomy," in *2019 American Control Conference (ACC)*. IEEE, 2019, pp. 2676-2682.
- [12] H. He, K. Wang, and R. Li, "To stay or to bypass: Unraveling mainline vehicles' aggregate strategic decision-making at highway weaving ramps," *arXiv preprint arXiv:2505.08965*, 2025.
- [13] R. Li, J. Liu, and R. Horowitz, "A game-theoretic model for aggregate lane choice behavior of highway mainline vehicles at the vicinity of on-ramps," in *2020 American Control Conference (ACC)*. IEEE, 2020, pp. 5376-5381.
- [14] R. Li, P. N. Brown, and R. Horowitz, "Employing altruistic vehicles at on-ramps to improve the social traffic conditions," in *2021 American Control Conference (ACC)*. IEEE, 2021, pp. 4547-4552.
- [15] E. Vinitsky, N. Lichtlé, K. Parvate, and A. Bayen, "Optimizing mixed autonomy traffic flow with decentralized autonomous vehicles and multi-agent reinforcement learning," *ACM Transactions on Cyber-Physical Systems*, vol. 7, no. 2, pp. 1-22, 2023.
- [16] M. J. Lighthill and G. B. Whitham, "On kinematic waves ii. a theory of traffic flow on long crowded roads," *Proceedings of the royal society of london. series a. mathematical and physical sciences*, vol. 229, no. 1178, pp. 317-345, 1955.
- [17] L. Muñoz, X. Sun, R. Horowitz, and L. Alvarez, "Traffic density estimation with the cell transmission model," in *Proceedings of the 2003 American Control Conference, 2003.*, vol. 5. IEEE, 2003, pp. 3750-3755.
- [18] C. F. Daganzo, "The cell transmission model: A dynamic representation of highway traffic consistent with the hydrodynamic theory," *Transportation research part B: methodological*, vol. 28, no. 4, pp. 269-287, 1994.
- [19] A. Pigou, *The economics of welfare*. Routledge, 2017.
- [20] C. Yu, A. Velu, E. Vinitsky, J. Gao, Y. Wang, A. Bayen, and Y. Wu, "The surprising effectiveness of ppo in cooperative multi-agent games," *Advances in neural information processing systems*, vol. 35, pp. 24 611-24 624, 2022.